NATURE OF MATTER
Elements, Compounds and Mixtures

Class 8 Science — Tuition Notes

Student-friendly notes based on the textbook pages provided.

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1. Learning Objectives

- Explain how matter is classified into pure substances and mixtures.
- Define elements, compounds, homogeneous mixtures and heterogeneous mixtures.
- Compare the two case studies from the textbook.
- Recall key historical thinkers and explain why water is a compound.

2. What Is Matter?

- Matter is anything that has mass and occupies space.
- Pure substance: a material made of one chemical substance, with a fixed composition and characteristic properties. Examples: pure copper, oxygen, distilled water.
- Mixture: two or more substances physically combined. The proportions can vary. Examples: air, seawater, soil and a snack mix.
- Important: A pure substance may be an element or a compound. Pure water is a compound, not an element.

3. Pure Substances

- A pure substance has a uniform composition and characteristic properties.
- It has definite melting and boiling points under a given pressure.
- It cannot be separated into its components by ordinary physical methods.
- It may be an element (such as copper) or a compound (such as water).

4. Elements

- An element is a pure substance made of only one kind of atom. It cannot be broken down into simpler substances by ordinary chemical methods.
- Examples and symbols: hydrogen (H), oxygen (O), sodium (Na), potassium (K), zinc (Zn), carbon (C), sulphur (S), mercury (Hg), calcium (Ca), iron (Fe).
- Symbol rule: the first letter is capital. If there is a second letter, it is lowercase: Co and CO mean different things.

5. Compounds

- A compound is a pure substance formed when two or more different elements chemically combine in a fixed proportion.
- Examples: water (H_2O), carbon dioxide (CO_2), sodium chloride (NaCl).
- A compound has properties different from those of its constituent elements.
- Its elements are present in a fixed ratio. Its components cannot be separated by simple physical methods; chemical processes are needed.

6. Mixtures

- A mixture forms when two or more substances are physically combined without forming a new substance.

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- The composition can vary, and the components generally retain their own properties.
- Mixtures can often be separated by physical methods such as handpicking, filtration, evaporation, sieving or using a magnet, depending on the materials.
- A mixture may be uniform or non-uniform.

7. Types of Mixtures

- Homogeneous mixture: has a uniform composition throughout; different components are not visibly distinguishable. Examples: salt solution, sugar solution, air, vinegar and brass.
- Heterogeneous mixture: does not have a uniform composition throughout; different components or regions may be distinguishable. Examples: snack mix, muddy water, soil and oil mixed with water.
- Solution: a homogeneous mixture, such as salt water.
- Alloy: a mixture containing a metal and one or more other elements, such as brass or bronze.
- Colloid: a mixture with tiny dispersed particles, such as milk or fog.
- Suspension: a heterogeneous mixture whose particles may settle on standing, such as muddy water.

8. Case Studies: Box 1 and Box 2

- Case Study 1 — Box 1: The crystals looked alike in shape and colour. Even after crushing, the smaller pieces looked similar. The story says samples from different places tasted the same. The intended conclusion is that Box 1 contains a pure substance.
- Case Study 2 — Box 2: The box contained rice puffs, peanuts, cornflakes, raisins and salt crystals. The parts looked different and could be separated by hand. The conclusion is that Box 2 contains a heterogeneous mixture.
- Safety: Never taste unknown substances in a classroom or laboratory. Use teacher-approved observation methods only.

9. Case Study Comparison

- Looking inside: Box 1 — similar-looking crystals; Box 2 — different snack ingredients.
- Checking appearance: Box 1 — crushed pieces still looked alike; Box 2 — ingredients had different colours and shapes.
- Tasting (as described in the story): Box 1 — samples described as tasting the same; Box 2 — components described as tasting different. This is a textbook story, not a safe classroom test.
- Trying to separate: Box 1 — no distinct components were seen to pick out; Box 2 — peanuts and raisins could be picked out by hand.
- Final conclusion: Box 1 — pure substance (the textbook's intended conclusion); Box 2 — heterogeneous mixture.

10. Pure Substance vs Mixture

- Pure substance: fixed composition; one element or one compound; characteristic melting/boiling points; not separated by ordinary physical methods.
- Mixture: variable composition; two or more substances physically combined; often changes state over a range; components can often be separated physically.

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- Examples of pure substances: pure copper and distilled water. Examples of mixtures: air, seawater and snack mix.

11. Early Ideas About Matter and Chemistry

- Empedocles: proposed four basic substances — earth, air, fire and water.
- Leucippus and Democritus: suggested matter is made of tiny particles. Democritus used the word “atomos”, meaning indivisible.
- Maharishi Kanada: associated with the ancient idea that matter is made of tiny particles.
- Antoine Lavoisier: helped establish modern chemistry through careful measurements and the Law of Conservation of Mass; published an important chemistry text in 1789.
- Joseph Proust: associated with the Law of Definite Proportions — a compound contains its elements in a fixed proportion by mass.
- John Dalton: developed an influential early modern atomic theory.

12. Water: Compound and Change of State

- Water is a compound because hydrogen and oxygen are chemically combined in a fixed ratio (H_2O). It is not simply a mixture of hydrogen and oxygen.
- When water freezes into ice or changes into water vapour, it remains H_2O. These are physical changes of state; the particles' arrangement and movement change.
- Passing electricity through water under suitable conditions can decompose it into hydrogen and oxygen gases. This process is called electrolysis.
- Safety: Electrolysis needs teacher supervision and suitable equipment. Never use a household power supply or an open flame near generated gases.

13. Quick Revision — Definitions

- Matter: anything that has mass and occupies space.
- Pure substance: material with fixed composition and characteristic properties, consisting of one element or one compound.
- Element: pure substance made of one kind of atom.
- Compound: pure substance formed when different elements chemically combine in a fixed proportion.
- Mixture: two or more substances physically combined in variable proportions.
- Homogeneous mixture: mixture with uniform composition throughout.
- Heterogeneous mixture: mixture with non-uniform composition.
- Solution: a homogeneous mixture. Alloy: a mixture containing a metal and other element(s).
- Colloid: mixture with tiny dispersed particles. Suspension: heterogeneous mixture whose particles may settle.

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14. Practice Questions

A. Very short answer

1. What is matter?
2. Define an element.
3. What is a compound? Give two examples.
4. What is a mixture?
5. Differentiate homogeneous and heterogeneous mixtures.
6. Why is water called a compound?
7. Name one method used to separate a mixture.
8. What is an alloy? Give one example.
9. What does “atomos” mean?
10. Who is associated with the Law of Definite Proportions?

B. Short answer

11. Write four properties of a pure substance.
12. Write four differences between a pure substance and a mixture.
13. Explain the observations and conclusions from Case Study 1 and Case Study 2.
14. Differentiate between an element and a compound with examples.
15. Why is freezing water into ice a physical change?
16. Write one contribution each of Democritus, Lavoisier, Proust and Dalton.

C. Classify each item

17. Classify each as element, compound, homogeneous mixture or heterogeneous mixture: oxygen, distilled water, air, brass, muddy water, sodium, sugar solution, rice puffs with peanuts and raisins, carbon dioxide, soil.

D. Think and explain

18. How would daily life change if water were a mixture of hydrogen and oxygen instead of a compound?
19. Why can components of a snack mix be separated by hand but the elements in water cannot?
20. Is every pure substance an element? Explain with an example.

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